

ORIGINAL ARTICLE

Risk-Factor Analysis of High School Basketball–Player Ankle Injuries: A Prospective Controlled Cohort Study Evaluating Postural Sway, Ankle Strength, and Flexibility

Hsing-Kuo Wang, PhD, Chia-Hong Chen, MS, Tzzy-Yuang Shiang, PhD, Mei-Hwa Jan, MS, Kwan-Hwa Lin, PhD

ABSTRACT. Wang H-K, Chen C-H, Shiang T-Y, Jan M-H, Lin K-H. Risk-factor analysis of high school basketball–player ankle injuries: a prospective controlled cohort study evaluating postural sway, ankle strength, and flexibility. *Arch Phys Med Rehabil* 2006;87:821-5.

Objective: To analyze risk factors, including postural sway, ankle strength, and flexibility, for the prediction of ankle injuries in men's high school basketball players.

Design: A cohort study with follow-up duration of 1 basketball season.

Setting: Biomechanics laboratory.

Participants: Forty-two (age, 16.5 ± 1.1 y) players competing in first league of the High School Basketball Association without history of injury in the lower extremities within 6 months before recruitment and without significant malalignment in the lower extremities were included. None of these players met exclusion criteria such as using ankle braces or taping or failed in wearing low-top sports shoes during the follow-up season.

Interventions: Not applicable.

Main Outcome Measures: Biomechanic measurements including isokinetic ankle strength, 1-leg standing postural sway, and ankle joint dorsiflexion flexibility were performed before the basketball season by 1 physical therapist. The subsequent monthly follow-up questionnaires were sent and returned by mail to prospectively record the incidence of ankle injury occurring in the season. Results of these preseason measurements were analyzed to correlate if any of these measured variables could predict future ankle injuries.

Results: Eighteen ankle sport injuries were recorded for 42 players during the follow-up season. High variation of postural sway in both anteroposterior and mediolateral directions corresponded to occurrences of ankle injuries ($P = .01$, odds ratio [OR] = 1.220; $P < .001$, OR = 1.216, respectively). All other variables were not associated with injury.

Conclusions: High variations of postural sway in 1-leg standing test could explain partly the increased prevalence of ankle injury in basketball players. It may be used as a screening tool to recommend balance training before basketball season.

Key Words: Athletic injuries; Biomechanics; Posture; Rehabilitation.

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ANKLE INJURIES IN BASKETBALL are believed to occur in a nonrandom manner and to be influenced by multiple factors, which have been classified as either intrinsic or extrinsic.¹ Prospective studies regarding basketball have been performed to identify intrinsic risk factors of ankle injury including unstable postural sway,² muscle weakness and imbalance,³ poor flexibility,³ hypermobile ankle joint,³ poor proprioception or joint sense,³ previous predisposing injury,⁴ and sex⁵ for the purpose of injury prevention. In addition, Milgrom et al⁶ and Beynon et al⁷ suggested intrinsic factors also include extreme body height and weight and anatomic malalignment of the ankle and foot because these variables (or factors) significantly increase risk of ankle injuries in athletes. For example, a very cavus foot structure would be a risk factor for lateral ankle sprain because this foot type usually is also associated with a hypomobile foot and reduced contact area and stability with the support surface. However, in these studies, the effects of intrinsic risk factors were measured in the absence of controls for extrinsic risk factors (confounding factors). The confounding factors of ankle injuries in athletes include shoe type,^{4,8} taping,⁹ or orthosis¹⁰ and playing surfaces,^{11,12} which were not accounted for in previous studies on this topic. Therefore, without well-controlled studies, little consensus has been reached regarding effects of intrinsic factors to ankle injury and the determination of priorities in risk reduction is not currently possible.

Our report describes a controlled cohort study that was performed to analyze the effects of intrinsic risk factors in ankle injury, including postural sway, strength, and ankle flexibility by a logistic regression model in men's high school basketball players. Possible confounding effects of other intrinsic factors (sex, injury history, foot types, malalignment) and extrinsic factors (shoes, ankle tape or brace, playing surfaces) on ankle injuries were minimized in this study.

METHODS

Participants

Our institutional review board approved this study. Members of 10 teams who registered and competed in the first league of the High School Basketball Association, Taiwan, Republic of China (ROC), were selected before the 2003 season as eligible subjects. The teams were selected because their schools have indoor basketball stadiums with standard sports floors (hard maple wood, 28×15 m) and no specifically designed prevention programs for ankle injury. Five teams with 70 players (age, 16.7 ± 1.2 y) were randomly chosen (block randomization, $n = 2$)

From School and Graduate Institute of Physical Therapy, College of Medicine, National Taiwan University, Taipei (Wang, Jan, Lin); Department of Health, Tao-Yuan General Hospital, Tao-Yuan City (Chen); and Institute of Sports Equipment Technology, Taipei Physical Education College, Taipei (Shiang), Taiwan.

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Reprint requests to Kwan-Hwa Lin, PhD, No. 1, Section 1, Jen-Ai Rd, School and Graduate Institute of Physical Therapy, College of Medicine, National Taiwan University, Taipei 100, Taiwan, ROC, e-mail: khlin@ntu.edu.tw.

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from these 10 teams. All 70 players, as well as their parents, provided informed consent before inclusion. The additional inclusion criteria of players were (1) no previous surgery on the lower extremities, (2) no history of injury with residual symptoms (pain, "giving-away" sensations, endurance loss) in the lower extremities within the 6 months before recruitment, (3) no evidence of a leg-length discrepancy (difference of distance from the anterior superior iliac spine to the superior surface of the most prominent aspect of the medial malleolus) of more than 1.5cm, (4) no genu varum with a tibiofemoral angle of less than 4° , (5) no functional overpronation of the foot arch (the angle formed between the distal medial malleolus, the navicular tuberosity, and the first metatarsal head $\leq 90^\circ$) or cavus foot (angle $= 180^\circ$), and (6) no calcaneal valgus or varus (the angle formed between alignment between posterior calcaneus and distal leg $> 8^\circ$ or $< 2^\circ$). Fourteen players failed to meet these additional inclusion criteria and were excluded. The assessments regarding items 3, 4, 5, and 6 were performed before the 2003 season by 1 physical therapist, and conducted on players in a standing position with a series of bilateral anthropometric^a and goniometric measurements. Nine of the 70 players were randomly chosen to undergo a repeat of the same protocol 24 hours after the first assessments with the same physiotherapist. Results of 2 sides (dominant and nondominant sides) were blindly pooled, and intratester reliability was established with intraclass correlation coefficients (ICCs). The ICCs of the leg-length, tibiofemoral angle, foot arch angle, and calcaneal angle measurement were .87, .81, .91, and .81, respectively. An accepted ICC was determined as greater than .80 at the outset of this study.

Fifty-six of the 70 players satisfied these inclusion criteria, and the initial biomechanic measurements of ankle flexibility, strength, and postural sway were subsequently performed by 1 physical therapist on these 56 players before the 2003 season (fig 1). Details regarding the biomechanic measurements are described later. Players were excluded from this study during the follow up season if they (1) used ankle braces or taping during training or competition without subclinical symptoms of pain or injuries ($n=3$), (2) were absent from training or competitions for more than 3 weeks for reasons other than injuries ($n=7$), (3) were unable to return follow-up questionnaires for 1 month ($n=3$), (4) participated in training courses in stadiums without maple floors, or (5) failed to wear their team's low-top sports shoes during training ($n=1$). With respect to item 5, it should be noted that 2 of the same pairs of low-top sports shoes without air cells in the heel^b were provided for this study by the High School Basketball Association, ROC. Based on these exclusion criteria, 14 players were excluded from this study. Forty-two players (age, 16.5 ± 1.1 y; body height, 179.5 ± 5.4 cm; weight, 80.3 ± 4.4 kg) were analyzed for risk factors during 1 basketball season (see fig 1).

Biomechanic Measurement

All measurements were performed by 1 experienced physical therapist at our biomechanic laboratory before the basketball session. Leg dominance was defined by players preferred to kick a ball. The dominant and nondominant legs were tested separately in an order of block randomization.

Isokinetic ankle strength. Players were evaluated in a supine position and with the hip and knee flexed at 80° by the inversion-eversion test by using the Cybex 6000.^c Each test was initiated with the ankle positioned at 0° of plantarflexion and performed within a comfortable range of motion (ROM). Both concentric and eccentric ankle inversion and eversion isokinetic strengths were measured at speeds of 60° and 180° /s. Aydog et al,¹³ by using the Biodex dynamometer, studied

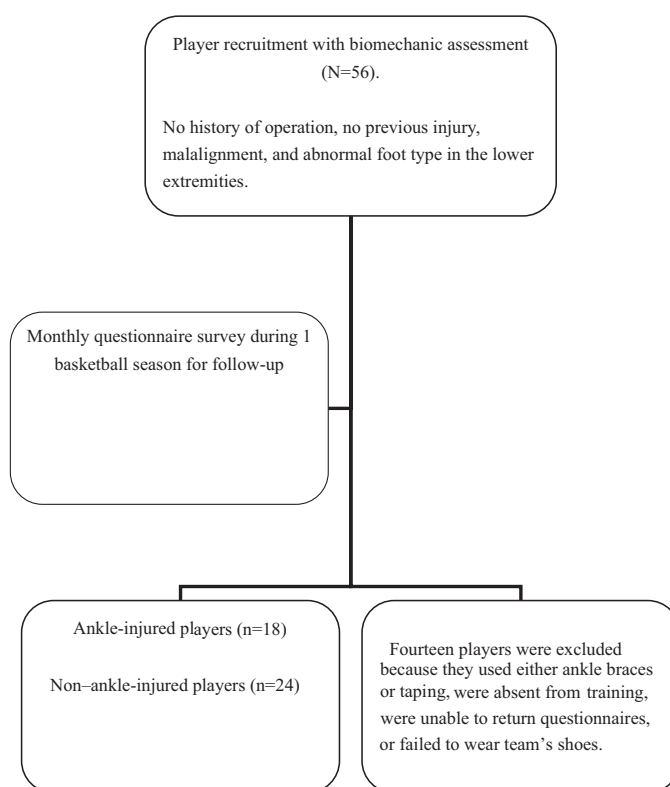


Fig 1. The flow chart of players in the follow-up after initial biomechanic assessments in posture sway, ankle strength, and flexibility.

test-retest reliability and found that the ICC values for strength of ankle invertor and evtor muscles at 60° and 180° /s angular velocities ranged from .81 to .96. The mean peak strength of each test was calculated from the 5 success reliable performances (the values for which did not vary by more than 10%) and was normalized according to the player's body weight. The strength ratio was defined as the ratio of the mean peak eversion strength to the mean peak inversion strength at the speed of 60° or 180° /s. In the endurance test, each player was asked to do 40 repetitions at his best concentric and eccentric strength of eversion and inversion. The endurance ratio was calculated as the ratio of the total work of the last 20 repetitions to that of the first 20 repetitions. Normalized strength, peak strength, and endurance ratios served as independent variables of ankle risk factors in this study.

Postural sway. Postural sway was assessed through open-eye 1-leg standing performance and was measured on the forceplate.^d To reduce noise, the forceplate was settled with a sampling rate of 600Hz, and high-pass and low-pass filters were set at 10.5 and 1500Hz, respectively. To get reliable results, 2 practice trials were permitted before measurements were performed. Players were asked to stand on each bare foot for at least 15 seconds with the other leg slightly flexed and each hand placed on the opposite shoulder (ie, across their chests). The first 5 seconds were not calculated to minimize interference from the initial preparation, and subsequent 10-second data were measured for postural sway analysis.¹⁴ All results of 1-leg standing performance were processed by Lab-View 6i graphic-based programming language.^e Variation of postural sway was quantified as twice the standard deviation (SD) of the averaged distance between the center of pressure

Table 1: Physical Characteristics and Training History of Injured and Uninjured Players

Players	n	Age (y)	Body Height (cm)	Body Weight (kg)	Training History (y)	Training (h/mo)
Uninjured players	24	16.5±1.1	179.0±5.5	71.0±8.9	3.8±1.3	120.8±16.3
Injured players	18	16.6±1.0	180.2±5.4	79.2±10.1	3.6±1.0	116.4±21.7
<i>P</i>		.801	.32	.268	.646	.704

NOTE. Values are n or mean ± standard error of the mean (SEM).

and reference point (ie, global coordinate zero point) on the forceplate in the mediolateral (ML) and anteroposterior (AP) axes during 10-second trials.¹⁵ Twice the SD covers 95% of the distribution of postural sway during 1-leg standing performance. Values in the ML and AP directions were both used as independent variables of ankle risk factors in this study.

Ankle ROM. Standard goniometer techniques were conducted on players in a prone position and used to measure the active range of ankle dorsiflexion with reported ICC ranging from .78 to .94 in both knees flexed to 90° and extended.¹⁶

Questionnaires

Two different questionnaires were used during the season. The first type was a recruitment questionnaire designed to record physical characteristics, play positions, training history, and training hours per month. The second type was a monthly follow-up questionnaire designed to prospectively record the incidence of injuries and profiles of ankle sports injury (location, time of occurrence) occurring in the basketball season. The recruitment questionnaires were mailed to the coach of each team at the beginning of the season and followed by the monthly follow-up questionnaires. The 42 players reported their ankle sports injuries on the monthly follow-up questionnaires if they had any in the season. The coaches collected and mailed the completed questionnaires to the authors and recorded participation time for players of each team on a weekly basis for the whole basketball season.

Ankle sports injury was defined in this study as an incident or period occurring in connection with basketball during training or competition that occurs at the ankle, handicaps the player during performance, or completely prevents the player from playing basketball or makes the player look for treatment to continue playing basketball. Two physiotherapists visited players and confirmed ankle injuries if players reported one. The physiotherapists determined injury severity by counting the time loss of participation in training or competition.

Statistical Analyses

A Mann-Whitney *U* test was performed for comparison of mean differences of physical characteristics (body height, body weight, age) and training (history, hours per month) between injured and noninjured players. Chi-square tests were used to examine correlations between leg dominance and prevalence of ankle injuries. Data were analyzed by the Bonferroni *t* method (multiple-comparison procedure) to compare differences of variables between those with and without ankle injuries. Prediction of variables on occurrences of ankle injury was analyzed by logistic regression. Data were analyzed by using the S-PLUS2000 Professional Release3.^f All analyses were performed in the null form, and α was set at .05.

RESULTS

The duration of the 2003 basketball season was estimated by coaches for each team, with a mean of 46.0 ± 2.5 weeks. Eighteen ankle sports injuries were recorded for 42 players during the follow-up season, and 11 occurred in the dominant

foot and 7 occurred in the nondominant foot. Fourteen and 4 injuries were categorized as mild and moderate degrees, respectively. Age, physical characteristics, training history, and training hours did not differ significantly between injured and uninjured players (all $P>.05$) (table 1). Effects of leg dominance in these injured players were not observed because the correlation between leg dominance and occurrence of ankle injury was not significant ($P>.05$). Comparisons of the injured and uninjured ankles showed that the differences of variation in magnitude of postural sway in the ML direction were statistically significant ($P<.001$) (table 2). In addition, players who displayed a high variation of postural sway in the ML or AP directions were likely to have an ankle injury during the basketball season in logistic regression tests ($P=.01$, odds ratio [OR]=1.220; $P<.001$, OR=1.216, respectively) (see table 2).

Table 2: Comparison of Variables (Intrinsic Risk Factors) Between Injured and Uninjured Ankles of Basketball Players

Variables	Injured Ankle (n=18)	Uninjured Ankle (n=66)	<i>P</i>
Standardized strength (ft-lb/lb of weight)			
60°/s concentric inversion	13.9±2.7	13.4±2.6	.55
180°/s concentric inversion	11.0±2.1	10.4±2.2	.32
60°/s concentric eversion	12.9±2.3	12.4±2.1	.37
180°/s concentric eversion	11.3±2.9	10.3±1.8	.28
60°/s eccentric inversion	13.7±4.0	15.0±3.5	.27
180°/s eccentric inversion	14.2±3.7	14.2±3.4	.96
60°/s eccentric eversion	14.2±3.1	15.0±3.1	.40
180°/s eccentric eversion	14.4±3.3	14.6±3.0	.84
Eversion/inversion ratio (%)			
60°/s concentric	93.2±14.5	94.8±16.7	.71
180°/s concentric	103.1±20.3	101.1±20.0	.74
60°/s eccentric	102.1±17.9	97.2±22.0	.45
180°/s eccentric	100.7±21.6	103.2±21.9	.68
Endurance index (%)			
Inversion	75.2±9.8	78.2±13.4	.44
Eversion	67.4±11.8	69.7±13.2	.50
Variation in postural sway (mm)			
ML axis	33.8±8.1	23.8±5.6	<.001*†
AP axis	23.2±5.3	19.6±3.7	.015†
Active ankle dorsiflexion (deg)			
Knee extended	6.6±4.0	8.7±3.8	.71
Knee flexed to 90°	17.4±3.0	18.7±5.3	.30

NOTE. Values are mean ± SEM.

*Significant differences between the 2 groups ($P<.05$) in Bonferroni *t* method.

†Significance in logistic regression (dependent variable: ankle injury); ML: odds ratio (OR)=1.216 (95% confidence interval [CI], 1.089–1.359; $P<.001$).

‡Significance in logistic regression (dependent variable: ankle injury); AP: OR=1.220 (95% CI, 1.046–1.424; $P=.01$).

DISCUSSION

Findings of this study reveal that only the players who displayed a high variation of postural sway were most likely to have an injured ankle during the basketball session (see table 2). Significant differences of the variation between the uninjured and injured ankles in the ML direction were found. This may be because postural sway represents the ability to maintain a standing balance, and a large variation of postural sway may indicate inconsistent or poor control of ankle stability. Basketball performance places a high demand on standing stability,¹⁷ and an abnormal or inconsistent ability to control postural sway in both AP and ML directions may indicate a functional instability and a poor quality of performance, leading to ankle injury. Similar findings regarding effects of postural sway on ankle injury have been made by other investigators^{2,18,19} in football and basketball players without controls for other risk factors. Football and basketball players with greater postural sway area and higher velocity of postural sway, respectively, were found to have more ankle injuries than other players during the competition season.^{2,18} From present results, the odds ratio of eyes-open 1-leg standing postural sway in injured versus uninjured athletes was around 1.2. Results of the eyes-open 1-leg standing performance test could only explain an increased prevalence of 20% and may not be used as an effective predictor of ankle sports injuries in these basketball players. However, this significant association with ankle injury suggests that 1-leg standing may be used as a screening tool for therapists or doctors to recommend balance training before basketball season. Findings of the present report are in agreement with other studies that observed no significant correlations between ankle isokinetic strength, dorsiflexion range, and ankle injuries.^{3,7,19}

In this study, confounding variables frequently ignored in previous studies were controlled at recruitment. These variables included predisposing injury,⁴ sex,⁵ foot type,^{4,7} malalignment in the lower extremity,⁷ footwear,⁸ ankle protective devices,^{10,11} and playing surfaces.^{12,13}

This study has several limitations. First, the basketball players in this study were recruited from a highly developed urban area and trained at a higher competition level with modest to high intensity. Therefore, the results may not be applicable to recreational basketball players or high school players in other settings. Second, we acknowledge that severe ankle or knee injuries occurring 6 months before the recruitment may still affect postural sway. For this study, we assumed that players who had injuries without any residual symptoms for 6 months did not differ significantly from players who had no history of injuries, although athletes might have subclinical dysfunctions from a prior injury. Third, postinjury examinations were not performed in these injured players. Previous studies have shown that (1) increased amplitudes of postural sway, (2) a decreased active ROM in eversion and inversion, (3) a deficit in evertor muscle peak torque, and (4) an evertor-invertor muscle imbalance were found on the injured ankle as compared with the uninjured side.²⁰ In this study, it is not clear whether ankle injuries further influenced these variables. Further studies are suggested to compare these variables before and after ankle injuries.

CONCLUSIONS

Results of the high postural sway in eyes-open 1-leg standing test before basketball session correlated significantly with subsequent ankle injuries, although it could explain only a slight increase of prevalence in the rate of ankle injury. How-

ever, we suggest that sports medicine personnel may use this 1-leg standing as a screening tool to recommend balance training before the basketball season. Further studies are suggested to compare postural sway variables before and after ankle injuries.

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